

Assignment 6

This exercise sheet is dedicated to numerical methods for the solution of nonlinear equations. Apart from Exercise 22, which is still for the scalar case, all other exercises are for systems of equations.

Graded exercise

Exercise 21 (Using Newton's method in optimization problems)

In this exercise we look at how we can use Newton's method to solve optimization problems. More precisely, the goal of this exercise is to find numerically the maximizer

$$x_* = \operatorname{argmax}_{x \in \mathbb{R}^2} f(x), \quad (1)$$

where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is given by

$$f(x) = \exp\left(-\frac{\|x\|^2}{2}\right), \quad (2)$$

and $\|\cdot\|$ denotes the Euclidean norm. If you attend the Statistics course, you can observe that f is, up to a constant, the density of a standard bivariate normal distribution, and such kind of optimization problems arise from maximum likelihood estimation.

A way to look for a maximum (or minimum) of a function is to search for the stationary points. This means that we search for x_* such that $\nabla f(x_*) = 0$. If the Hessian of f is positive (resp. negative) definite at x_* , then we know that x_* is a maximum (resp. minimum).

In this simple example, we know a priori that f has a unique maximum at the origin, and this is the only location where ∇f vanishes. Solving (3) is therefore equivalent to computing $x_* \in \mathbb{R}^2$ such that $\nabla f(x_*) = 0$.

To find the numerical solution, we define $F := \nabla f$ and apply Newton's method to find the root of F .

- (a) For f as in (2), compute $F = \nabla f$ and $JF = \text{Hess}(f)$ by hand. Here $\text{Hess}(f)$ stands for the Hessian matrix of f .

- (b) Write a PYTHON function `Newton` which, given in input a function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and its Jacobian matrix JF , uses Newton's method to find (approximately) a root of F .

In input, `Newton` should take: F and JF as lambda functions, an initial guess x_0 , the maximum number of iterations allowed and a tolerance for the absolute error between two consecutive iterates, measured in the Euclidean norm.

In output, `Newton` should return the approximate root, the total number of iterations and an array with the absolute errors of all iterates.

Note that for the moment F is a generic function, later we will call the routine for our specific case of $F = \nabla f$. Note also that there should be no difference in coding the method on $\mathbb{R}^2$ or in general dimension $\mathbb{R}^m$.

Hint: To solve linear systems, you can use the `numpy` routine `linalg.solve`, and to measure distances you can use `linalg.norm`.

- (c) Write a PYTHON script that uses the routine from (b) and the computations from (a) to calculate the maximum of f via finding the zero of $F = \nabla f$. In the same script, produce a graph of absolute error of the iterates versus the iteration number (to be saved and handed in separately in the assignment).

When applying Newton's method, you can use the initial guess $x_0 = (0.25, 0.25)^\top$, an absolute error tolerance of 10^{-8} and an upper bound of 1000 on the number of iterations.

We now observe that the location of the maximum is invariant to monotone transformations of f . More precisely, we can find x_* as in (3) by solving

$$x_* = \operatorname{argmax}_{x \in \mathbb{R}^2} \log f(x). \quad (3)$$

If you follow the Statistics course, $\log f$ would be the analogous of a log-likelihood.

- (d) Repeat task (c) but now for $\log f$ instead of f , with f as in (2). In how many iterations does Newton's method converge? How do you explain this?

Remark: It is not part of the task, but one can also see that in this case Newton's method converges globally.

Other exercises

Exercise 22 (Modified Newton's method)

In class we have always assumed that the root of the function f we were looking for had multiplicity one, that is, it was a simple root. In this exercise, we look at how we can modify Newton's method to find a root with multiplicity $m > 1$, in case the

multiplicity is known a priori. If the multiplicity is not known a priori, then other versions of Newton's method are possible, such as the adaptive Newton algorithm (that we do not address here).

We have seen in class that Newton's method can be seen as a fixed point iteration for the function

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

If x_* is a simple root, then $g'(x_*) = 0$ and $g''(x_*) = \frac{f''(x_*)}{f'(x_*)}$, and these relationships ensure second order convergence of the method, thanks to the following result, also seen in class.

Proposition *If $g \in C^{p+1}(\bar{I})$ for an open neighborhood I of the fixed point x_* and an integer $p \geq 1$, and if $g^{(i)}(x_*) = 0$ for $1 \leq i \leq p$ and $g^{(p+1)}(x_*) \neq 0$, then the fixed-point method with iteration function g has order $p + 1$.*

If instead x_* has multiplicity $m > 1$, then Newton's method is no longer second-order convergent.

- (a) Show that, indeed, if x_* has multiplicity $m > 1$ and f is m -times differentiable ($m \geq 1$), then $g'(x_*) = 1 - \frac{1}{m} \neq 0$.

Hint: The fact that x_* is a root of multiplicity m means that $f(x_*) = f'(x_*) = \dots = f^{(m-1)}(x_*) = 0$ and $f^{(m)}(x_*) \neq 0$. Set then $f(x) = (x - x_*)^m h(x)$, h being a function such that $h(x_*) \neq 0$.

- (b) If the value of m is known a priori, then the quadratic convergence of Newton's method can be recovered by resorting to the so-called *modified Newton's method*

$$x_{k+1} = x_k - m \frac{f(x_k)}{f'(x_k)}, \quad k \geq 0.$$

Under the assumptions of task (a), show that the modified Newton method has order of convergence equal to 2.

Exercise 23 (Convergence of fixed point iteration)

Consider the mapping $G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with components

$$G_1(u) = \cos\left(\frac{1}{2}u_1 + \frac{1}{4}u_2\right), \quad G_2(u) = \cos\left(\frac{1}{3}u_1 u_2\right).$$

- (a) Explain why it is obvious that any fixed point u_* of G must lie in the set $\mathcal{D} = [-1, 1] \times [-1, 1]$.
- (b) Show that G is a contraction mapping on $\mathcal{D}$ with respect to the maximum norm, with Lipschitz constant $L = \frac{3}{4}$.
- (c) Show that G has a unique fixed point $u_* \in \mathcal{D}$.

- (d) Why does fixed point iteration converge to u_* for every starting point $u_0 \in \mathbb{R}^2$?

Exercise 24 (Convergence of Newton's iteration)

Let

$$F(u) = \begin{pmatrix} u_1^2 + 4u_1u_2 - 3 \\ u_1 - u_2^2 - 3 \end{pmatrix}, \quad \text{for } u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \in \mathbb{R}^2.$$

- (a) Show that the first iterate when using Newton's method is $(3, -\frac{1}{2})^T$ if we choose $(3, 0)^T$ as initial guess.
- (b) Show that Newton's method breaks down if we choose $(9, -3)^T$ as initial guess.